

Department of Artificial Intelligence and Machine Learning

IntelliData — An Agentic AI Based Data Intelligence & Governance Platform

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Problem Statement and Motivation

- ❑ Real-world datasets are messy: **missing values, duplicates, outliers, and inconsistent types** must be handled before any analysis or ML.
- ❑ **Manual profiling, cleaning, and EDA is repetitive**, time-consuming, and hard to reproduce or audit.
- ❑ **Fully automated tools and LLM assistants** may silently change data, claim causation from correlation, or send raw records to external APIs, with no human control or traceability.
- ❑ **Need:** A governed, agent-orchestrated platform that profiles, scores, cleans, and analyzes data, generates AI insights, and keeps **humans in control** with versioning and a full audit trail.

User and Stakeholder Analysis

- ❑ **Data Analysts / Scientists** – Quickly profile, clean, and explore datasets through explainable, reviewable steps.
- ❑ **Business Users / Decision Makers** – Get plain-language summaries and hypotheses without writing code.
- ❑ **Students / Researchers** – Learn structured, reproducible data preparation with full traceability.

Expected Outcome

- ❑ An analysis-ready, versioned dataset with quality scores, EDA charts, ML-readiness diagnosis, human-reviewed AI insights, and an HTML governance report with the complete audit trail.

Proposed agentic AI solution

IntelliData is a **multi-agent, governed AI platform** that automates data intelligence workflows.

- ❑ **Supervisor Agent** – Orchestrates all agent transitions and halts at human approval checkpoints.
- ❑ **Intake Agent** – Inspects schema, empty columns, identifier columns, and date-like strings (read-only).
- ❑ **Quality Agent** – Scores completeness, duplicates, type consistency, and IQR outliers (0–100).
- ❑ **Cleaning Planner Agent** – Proposes risk-rated (low/medium/high) cleaning actions; never auto-executes.
- ❑ **Cleaning Executor Agent** – Applies only approved proposals and commits new dataset versions.
- ❑ **Visualization Agent** – Recommends semantically safe charts (never sums ratings, %, or IDs).
- ❑ **ML Readiness Agent** – Checks target leakage, class imbalance, scaling, and encoding needs.
- ❑ **Insight Agent** – Generates hypotheses with confidence levels; no causal claims.
- ❑ **Report Agent** – Compiles findings and audit trail into an HTML governance report.

Dataset and knowledge source identification

❑ User-Uploaded Datasets

Source: CSV / Excel files uploaded through the Streamlit UI

Validated for file extension and size; sample workspace uses Titanic-Dataset.csv

❑ Derived Statistical Knowledge

Column profiles, cardinality, missing %, quantiles, IQR outliers

Distributions and Pearson correlations for EDA and AI prompts

❑ Governance Knowledge

Approval decisions and cleaning proposals with risk levels

Dataset version graph (v0, v1, v2 ...) and immutable audit events

Selection of tools and Framework

- ❑ **Application & Language**

- ❑ Python 3.12 – Core language for agents, workflow, and tools
- ❑ Streamlit – Multi-page UI, Agent Governance Hub, approval controls

- ❑ **Data Processing & ML**

- ❑ Pandas – Profiling, quality checks, versioned transformations
- ❑ scikit-learn – ML readiness diagnostics and baseline suggestions

- ❑ **AI Model**

- ❑ Google Gemini API (optional) – Executive summaries and structured hypotheses
- ❑ Local deterministic fallback – Pearson-based insights when no API key is available

Selection of tools and Framework

- ❑ **Orchestration & Governance**

- ❑ Supervisor + specialized agents – Workflow state machine with approval checkpoints
- ❑ Tool Registry – Only registered, deterministic tools; unknown tools rejected

- ❑ **Visualization**

- ❑ Plotly – Interactive distributions, correlations, and recommended charts

- ❑ **Testing & Deployment**

- ❑ pytest + Ruff – 20 governance tests, linting, dependency integrity check
- ❑ Docker + GitHub Actions – Containerized deployment and CI workflow

Agent workflow and Orchestration

- ❑ **Dataset Upload** – User uploads a CSV/Excel file or opens the sample workspace
- ❑ **Supervisor Launch** – Runs the autonomous pipeline with a max-step limit to prevent loops
- ❑ **Intake & Quality** – Schema inspection, then quality scoring (completeness, duplicates, outliers)
- ❑ **Cleaning Planning** – Risk-rated proposals created; workflow pauses for approval
- ❑ **Human Approval** – User approves, edits, or rejects each cleaning proposal
- ❑ **Execution** – Executor applies only approved proposals and creates version v1, v2 ...
- ❑ **Downstream Intelligence** – Visualization, ML readiness, then AI business insights
- ❑ **Review & Export** – Human reviews insights, authorizes export, Report Agent packages results

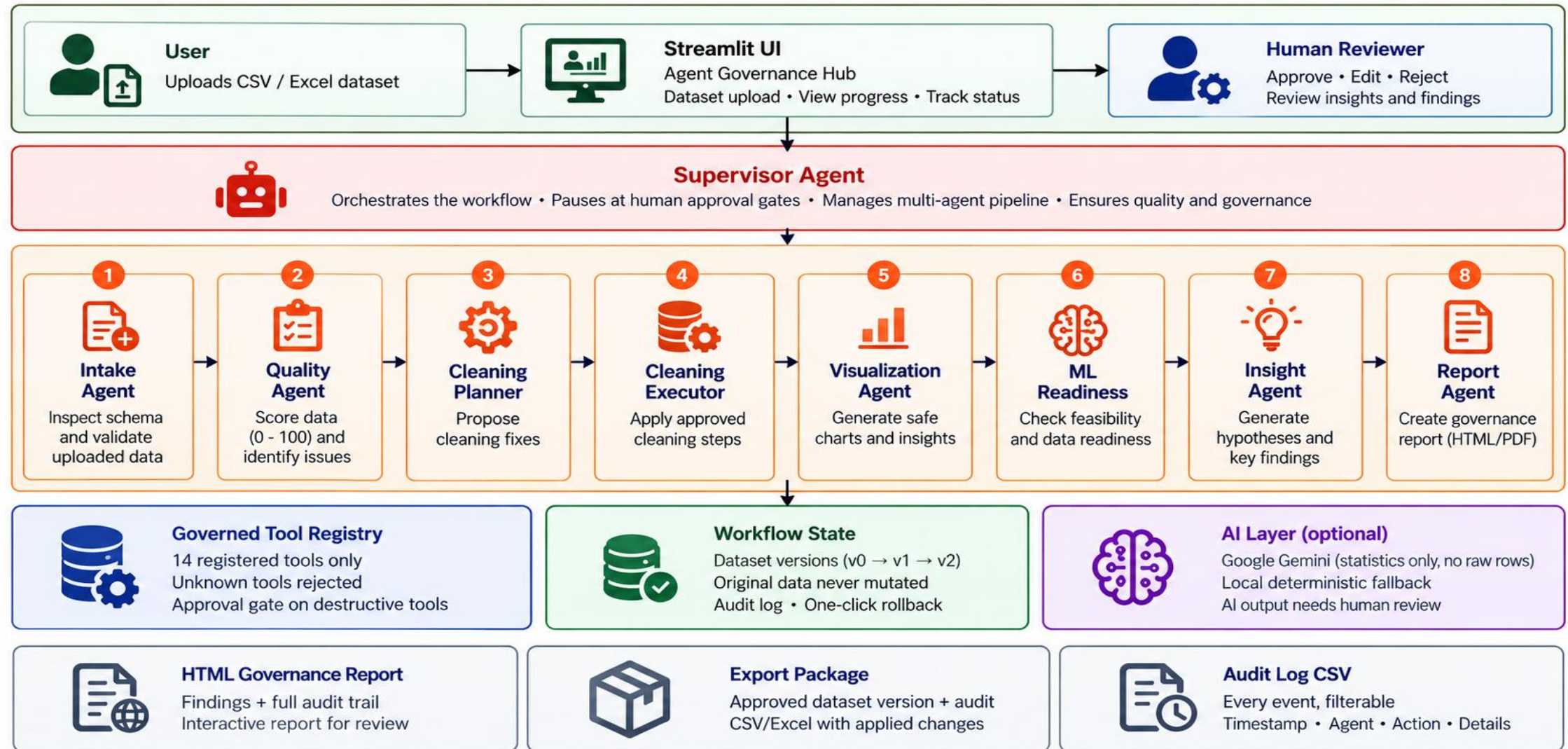
Governed Tool Registry

- ❑ **Registered Tools** – 14 tools (e.g., `validate_dataset`, `assess_quality`, `run_eda`, `execute_cleaning`)
- ❑ **Unknown Tool Rejection** – Unregistered names raise `UnregisteredToolError`
- ❑ **No Arbitrary Code** – Agents never execute LLM-generated code
- ❑ **Approval Gate** – `execute_cleaning`, `rollback_cleaning`, and `export_dataset` require approval
- ❑ **Preview Before Execute** – `preview_cleaning` runs in a sandbox before any change
- ❑ **Privacy Boundary** – Gemini receives only bounded statistics, zero raw rows; inputs sanitized
- ❑ **Graceful Fallback** – Local deterministic mode when `GEMINI_API_KEY` is absent

Human-in-the-Loop (HITL)

- ❑ **AI Proposals** – Planner creates proposals with risk levels; execution blocked while pending
- ❑ **Human Review** – User inspects each proposal with a "Why am I seeing this?" explanation
- ❑ **Approve** – Accept individually or via batch selection (no blind approve-all)
- ❑ **Edit & Approve** – Adjust parameters in a JSON editor (e.g., custom fill value)
- ❑ **Reject** – Original data preserved; optional note recorded in the audit log
- ❑ **High-Risk Guard** – Outlier capping and column drops need explicit confirmation
- ❑ **Audit & Rollback** – Every event logged immutably; one-click rollback to parent version
- ❑ **Export Authorization** – Report package generated only after final human approval

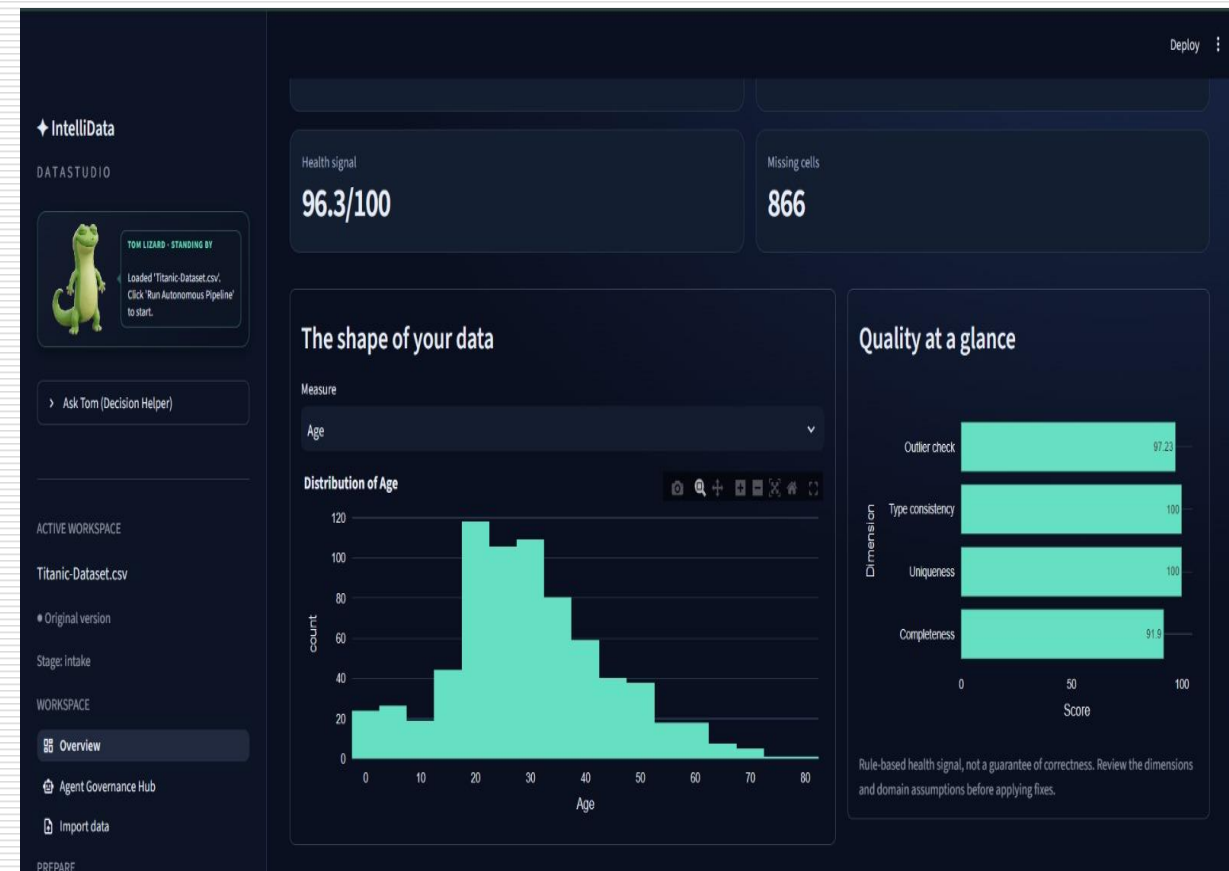
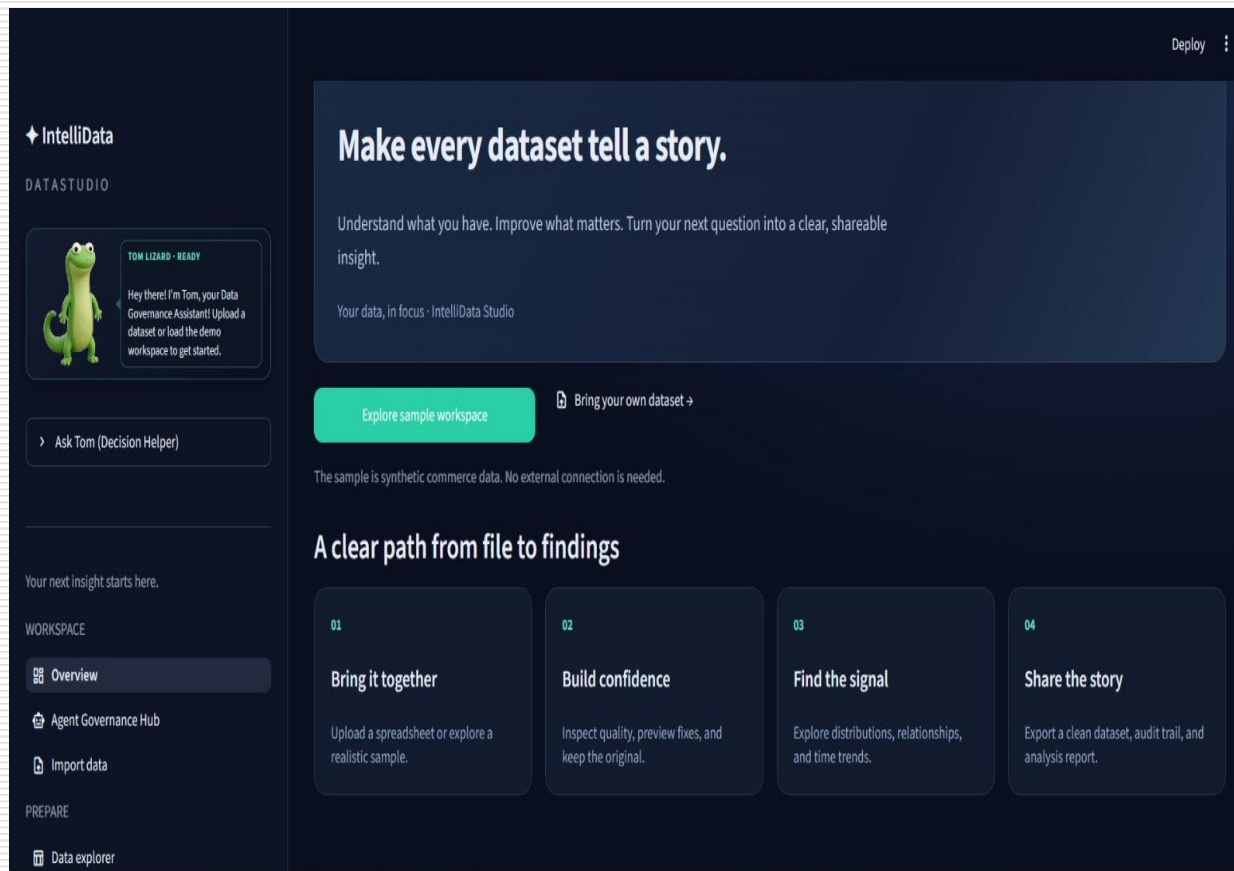
Architecture Diagram



Project plan and milestones

- ❑ **Phase 1–2:** Project setup, config, logging; dataset upload & validation
- ❑ **Phase 3–4:** Data profiling and data quality scoring
- ❑ **Phase 5:** Intelligent cleaning engine
- ❑ **Phase 6–7:** Exploratory data analysis and visualization dashboard
- ❑ **Phase 8–9:** Gemini AI service layer, AI cleaning explanations, executive summary
- ❑ **Phase 10–11:** AI business insights, feature engineering, ML readiness & recommendations
- ❑ **Phase 12–13:** AI dataset chatbot; PDF/HTML report generation
- ❑ **Phase 14–15:** UI polish, deployment, and portfolio wrap-up

Dashboard and Results



Dashboard and Results

The screenshot shows the 'Agent Orchestration Hub' interface. On the left is a sidebar with 'IntelliData' and 'DATASTUDIO' logos, a 'TOM LIZARD' character, and navigation links like 'Ask Tom (Decision Helper)', 'ACTIVE WORKSPACE', and 'WORKSPACE'. The main area displays 'Titanic-Dataset.csv - Original version - 891 rows'. It includes a 'GOVERNED AGENT WORKFLOW' section with a 'Data Intake' status of 'Idle', an 'ACTIVE VERSION' of 'v0 (1 version(s) tracked)', and 'STEPS' of '0 / 50'. Below this is an 'Agent Activity Timeline' with buttons for 'Data Intake' (Active), 'Data Quality' (Pending), 'Cleaning Planner' (Pending), 'Cleaning Approval' (Pending), 'Cleaning Executor' (Pending), 'Visualization' (Pending), 'ML Readiness' (Pending), 'Insight' (Pending), 'Insight Review' (Pending), 'Report' (Pending), and 'Export Authorist...' (Pending). At the bottom, there are buttons for 'Run Autonomous Pipeline', 'Step Once', and 'Reset Workflow', along with dataset information: 'Original dataset: Titanic-Dataset.csv (891 rows)' and 'Active version: v0 (891 rows)'. A 'Goverance & Approval Decisions (0 Pending)' section at the bottom indicates 'No approval requests have been submitted yet. Run the agent workflow to inspect proposals.'

This screenshot shows the 'Governance & Approval Decisions (0 Pending)' section. It features tabs for 'Governance & Approvals', 'Intelligence & Findings', and 'Full Audit Trail & Rollback'. Under the 'Resolved Approvals History (9)' tab, a list of nine approved actions is shown, each with the text 'APPROVED by human_user at [timestamp]'. The actions are: 'Protect Identifier Columns', 'Impute Missing Numeric: Age', 'Impute Missing Categorical: Cabin', 'Impute Missing Categorical: Embarked', 'Cap Extreme Outliers: Age', 'Cap Extreme Outliers: SibSp', 'Cap Extreme Outliers: Fare', 'Review AI Insights & Hypotheses', and another 'Review AI Insights & Hypotheses' entry.

Dashboard and Results

02 / Understand - Inspect individual columns, find records, and explicitly convert text dates before exploring trends.

Titanic-Dataset.csv - Cleaned version - 891 rows

Record explorer Column intelligence

Search records

Search any column...

Visible columns

PassengerId x Survived x Pclass x Name x Sex x Age x SibSp x Parch x Ticket x Fare x Cabin x Embarked x

891 matching rows. Displaying up to 1,000; exports include all matches.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	B96 B98	S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	65.6344	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	B96 B98	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	B96 B98	S
6	0	3	Moran, Mr. James	male	28	0	0	330877	8.4583	B96 B98	Q
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2.5	2.5	1	349909	21.075	B96 B98	S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333	B96 B98	S
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.0708	B96 B98	C

04 / Discover - Build an interactive chart, compare groups, and follow changes over time.

Titanic-Dataset.csv - Cleaned version - 891 rows

Visualization

Correlation

Filter by category

No filter

891 records in this view

Chart title

Give your chart a story

Filters only affect this chart. They do not modify your dataset.

Numeric columns

PassengerId x Survived x Pclass x Age x SibSp x Parch x Fare x

Method

pearson spearman

Correlation measures association; it does not establish causation.

Correlation

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
PassengerId	1.00	-0.01	-0.04	0.04	-0.07	0.00	0.00
Survived	-0.01	1.00	-0.34	-0.08	0.03	0.08	0.32
Pclass	-0.04	-0.34	1.00	-0.14	0.02	0.02	-0.72
Age	0.04	-0.08	-0.14	1.00	-0.25	-0.10	0.14
SibSp	-0.07	0.03	0.02	-0.25	1.00	0.41	0.35
Parch	0.00	0.08	0.02	-0.10	0.41	1.00	0.29
Fare	0.00	0.32	-0.72	0.14	0.35	0.29	1.00

Download interactive chart

Thank You